Dataset Implementation and Usage Guide

Intelligent and Secure Task Scheduling for Digital Twin-Enabled Edge-Cloud Networks

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All record counts, class distributions and storage figures in this document are computed from the actual files on disk. No values are fabricated.

1. Dataset Purpose

This project trains AI models and runs a security-aware Genetic Algorithm scheduler on publicly available real network-traffic datasets. Two clearly separated dataset categories are maintained: ATTACKED (malicious traffic with attack labels) and BENIGN (normal traffic with no attack labels). The AI risk model and GA fitness function both consume features derived from these datasets.

2. Why Datasets Are Required

The project's security risk prediction model (Random Forest classifier) must learn to distinguish benign from malicious activity. The Genetic Algorithm's fitness function incorporates an 'attack probability' derived from network-flow features; this requires trained models and real feature distributions, not synthetic data alone. Public datasets provide ground-truth labels and realistic attack patterns that synthetic data cannot reproduce.

3. Attacked Dataset: NSL-KDD (Local) and UNSW-NB15 (Manual)

The primary real attacked dataset used in this project is **NSL-KDD** (already imported locally in datasets/real/). The secondary/preferred attacked dataset is **UNSW-NB15**, which has 9 attack families and 2,540,044 records. UNSW-NB15 requires manual download from the official UNSW SharePoint.

NSL-KDD attack classes (39 distinct labels, computed at runtime):

Attack Class	Count	Percentage
neptune	45716	64.44%
satan	4360	6.15%
ipsweep	3643	5.14%
smurf	3108	4.38%
portsweep	3070	4.33%
nmap	1566	2.21%
back	1300	1.83%
guess_passwd	1284	1.81%
mscan	996	1.4%
warezmaster	964	1.36%
teardrop	901	1.27%
warezclient	890	1.25%
apache2	737	1.04%
processtable	685	0.97%
snmpguess	331	0.47%
saint	319	0.45%
mailbomb	293	0.41%
pod	221	0.31%
snmpgetattack	178	0.25%
httptunnel	133	0.19%
buffer_overflow	50	0.07%
multihop	25	0.04%
rootkit	23	0.03%
land	22	0.03%
named	17	0.02%
ps	15	0.02%
sendmail	14	0.02%
xterm	13	0.02%

imap	12	0.02%
ftp_write	11	0.02%
loadmodule	11	0.02%
xlock	9	0.01%
phf	6	0.01%
perl	5	0.01%
xsnoop	4	0.01%
spy	2	0.0%
worm	2	0.0%
sqlattack	2	0.0%
udpstorm	2	0.0%

NSL-KDD processed split (real numbers from pipeline): Attacked rows = 70,940 | Benign rows = 76,967 | Duplicates removed = 610 | Total = 147,907

UNSW-NB15 (manual download):

Official page: https://research.unsw.edu.au/projects/unsw-nb15-dataset
Download (login required): https://unsw-my.sharepoint.com/:f:/g/personal/z5025758_ad_unsw_edu_au/EnuQZZn3XuNBjgfcUu4DIVMBLCHyoLHqOswirpOQifr1ag
DOI: 10.21227/8vf7-s525
Records: 2,540,044 | Features: 49 + attack_cat + Label
Attack classes: Generic, Exploits, Fuzzers, DoS, Reconnaissance, Analysis, Backdoors, Shellcode, Worms
Benign: Yes (Normal class, approx. 2,218,761 records per published literature)

4. Benign Dataset

The benign dataset is the 'normal' class extracted from NSL-KDD. Original label: 'normal'. Filtered label: 'benign'. Records: 76,967 (computed from actual file). Source notes: Filtered from NSL-KDD: original label 'normal' -> unified label 'benign'. Zero duplicates, zero missing values in the first chunk of the processed file.

UNSW-NB15 Normal class (once downloaded): Approx. 2,218,761 benign records will be available by filtering Label==0 and attack_cat=='Normal'.

5. Dataset Source Table

Dataset	Type	Official Source	Format	Records	Purpose
NSL-KDD	ATTACKED + BENIGN	unb.ca/cic/datasets/nsl.html	CSV	148,517	Primary real IDS data
UNSW-NB15	ATTACKED + BENIGN	research.unsw.edu.au/projects/unsw-nb15-dataset	CSV/ARFF	2,540,044	Modern IDS benchmark
CIC-IDS2017	ATTACKED	unb.ca/cic/datasets/ids-2017.html	CSV/PCAP	2,886,020	Reference
CSE-CIC-IDS2018	ATTACKED	unb.ca/cic/datasets/ids-2018.html	CSV	16,800,000	Reference
TON_IoT	ATTACKED + BENIGN	research.unsw.edu.au/projects/toniot-datasets	CSV	461,043	Edge/IoT backup
BoT-IoT	ATTACKED + BENIGN	research.unsw.edu.au/projects/bot-iot-dataset	CSV/PCAP	73,370,044	Reference

6. Download Procedure

NSL-KDD: Already imported locally at datasets/real/. Run:
python scripts/download_datasets.py NSL-KDD

UNSW-NB15: Manual download required. Steps:
1. Open https://research.unsw.edu.au/projects/unsw-nb15-dataset
2. Click the SharePoint download link on the official page
3. Download UNSW-NB15_1.csv through UNSW-NB15_4.csv
4. Place files in datasets/raw/attacked/
5. Run: python scripts/validate_datasets.py

7. File Structure

datasets/
  raw/attacked/ -- original attacked downloads (immutable)
  raw/benign/ -- original benign downloads (immutable)
  real/ -- imported public dataset CSVs (NSL-KDD)
  processed/attacked/ -- cleaned attacked files for AI
  processed/benign/ -- cleaned benign files for AI
  metadata/ -- one JSON per dataset (source, checksums, license)
  statistics/ -- validation reports, class distributions

8. Dataset Size and Record Counts (actual)

File	Size (MB)	Rows
nsl_kdd_combined.csv	19.95	148,517
nsl_kdd_train.csv	16.90	125,973
nsl_kdd_test.csv	3.05	22,544
nsl_kdd_attacked.csv (processed)	computed	70,940
nsl_kdd_benign.csv (processed)	computed	76,967
nsl_kdd_features.csv (preprocessed)	computed	147,907

9. Features

NSL-KDD: 41 original flow features + label + difficulty column
Categorical: protocol_type, service, flag
Numeric: duration, src_bytes, dst_bytes, count, srv_count, rates, host stats, etc.

UNSW-NB15: 49 features (basic, content, time, general, connection)
Categorical: proto, service, state
Numeric: dur, spkts, dpkts, sbytes, dbytes, rate, sload, dload, etc.

Unified schema (project): timestamp, source, destination, protocol, source_port, destination_port, packet_count, byte_count, duration, flow_rate, network_latency, cpu_usage, memory_usage, task_size, security_label, attack_type, risk_score
Derived fields: flow_rate = src_bytes/(duration+1), byte_count = src_bytes+dst_bytes, risk_score = attack_type_to_risk(attack_type)

10. Train/Test Preparation

NSL-KDD: 80/20 stratified split (118,325 train / 29,582 test)
UNSW-NB15: Official split shipped (175,341 train / 82,332 test)
All preprocessing is done by: python manage.py preprocess_datasets

11. Preprocessing

Operations performed (in order):
1. Load CSVs (chunked, 200k rows at a time for memory efficiency)
2. Strip and normalise labels (case, trailing dots)
3. Remove exact duplicate rows
4. Separate into attacked/benign partitions
5. Label-encode categoricals (protocol_type, service, flag)
6. MinMax-normalise numeric features
7. Stratified 80/20 train/test split
8. Write outputs to datasets/processed/

12. Feature Engineering

Derived features added to processed CSVs:
risk_score: mapping from attack_type via ATTACK_RISK_MAP in data_pipeline/feature_engineering.py
flow_rate: src_bytes / (duration + 1)
byte_count: src_bytes + dst_bytes
These fields are documented and can be dropped or overridden by downstream models.

13. AI Model Usage

The preprocessed train/test CSVs feed:
- Random Forest classifier (engine/ai_engine.py, models/risk_model.pkl)
- Label: security_label_binary (0=benign, 1=attacked)
- Features: all 41 NSL-KDD features (encoded/normalised) + derived fields
- Training: python manage.py run_analysis (or scripts/train_ai.py)

14. Security Risk Calculation

attack_probability = RF.predict_proba(features)[:,1]
risk_score = ATTACK_RISK_MAP[attack_type] (feature_engineering.py)
SecurityRisk = ambient_risk + attack_contribution (engine/risk_engine.py)

15. MEC Scheduling Usage

The GA fitness function in engine/genetic_algorithm.py incorporates:
- security (30% weight): penalty = attack_probability * task.security_sensitivity
- privacy (20% weight): loss = privacy_loss_factor * task.privacy_sensitivity
- latency (25% weight): actual_network_latency
- energy (15% weight): energy_rate * task.cpu_cycles_giga
- deadline (10% weight): missed_deadline_penalty
The risk model's predictions directly influence which MEC node the GA selects for a task.

16. Google Drive Storage

Google Drive integration is provided by cloud/models.py (GoogleDriveFile, CloudFolder).
After validation, processed datasets can be uploaded via the Dataset Manager dashboard.
The Google Drive file ID is stored in the MySQL Dataset model.
Folder structure: Intelligent_Secure_Scheduling/datasets/attacked/, benign/, raw/, processed/
No Google credentials are exposed; the service account key is stored in .env only.

17. MySQL Metadata

Model: datasets.Dataset (datasets/models.py)
Fields: dataset_id, name, dataset_type, source, file_name, file_size, record_count, format, storage_location, google_drive_file_id, processing_status, validation_status, created_at, updated_at
Populated by: python manage.py dataset_statistics
Only metadata is stored in MySQL; raw dataset files are never put into the database.

18. Large Dataset Processing

All modules use chunked CSV reading (pd.read_csv with chunksize=200000).
No file is fully loaded into RAM if it exceeds ~2M rows.
Supported volumes: 100K, 500K, 1M, 5M+ rows (depending on source dataset size).
NSL-KDD on disk: 148,517 rows. UNSW-NB15 if downloaded: 2,540,044 rows.

19. Pipeline Commands

python scripts/download_datasets.py -- download/document official sources
python scripts/validate_datasets.py -- validate local files, write reports
python manage.py download_datasets -- Django command wrapper
python manage.py validate_datasets -- Django command wrapper
python manage.py preprocess_datasets -- clean + preprocess + feature engineer
python manage.py preprocess_datasets --force -- re-run cleaning
python manage.py dataset_statistics -- compute stats + sync MySQL

20. Data Flow Diagram

PUBLIC DATASET
|-- DOWNLOAD (or document manual source)
v
RAW DATA (datasets/raw/ or datasets/real/)
|-- VALIDATE (checksum, row count, columns, labels)
v
CLEANING (dedup, invalid row removal, label normalisation)
|-- ATTACKED / BENIGN split (processed/attacked/, processed/benign/)
v
PREPROCESSING (encode categoricals, normalise numerics, train/test split)
|-- FEATURE ENGINEERING (risk_score, flow_rate, byte_count)
v
AI MODEL (Random Forest, models/risk_model.pkl)
|-- SECURITY RISK (attack_probability per task)
v
GENETIC ALGORITHM (engine/genetic_algorithm.py)
|-- TASK SCHEDULING (MEC / Cloud node selection)
v
MEC / CLOUD EXECUTION
|-- RESULT
v
GOOGLE DRIVE (cloud/models.py integration)

21. License and Ethics

NSL-KDD: Public research benchmark. Cite: Tavallaee et al. 2009, 'A Detailed Analysis of the KDD CUP 99 Data Set'. See https://www.unb.ca/cic/datasets/nsl.html

UNSW-NB15: Free use for academic research (perpetuity). Commercial use must be agreed with the authors. Must cite: Moustafa & Slay 2015 (MilCIS), Moustafa & Slay 2016 (ISJ), Moustafa et al. 2017 (TBD), Moustafa et al. 2017 (DADS), Sarhan et al. 2020 (BDTA).

CIC-IDS2017/2018, TON_IoT, BoT-IoT: Research licence - see individual official pages.

The final deliverable ZIP contains dataset metadata and download instructions, not the raw datasets themselves (where the source licence prohibits redistribution).

22. Troubleshooting

ERROR: 'No such file' when running preprocess -> run download_datasets.py and/or check that datasets/real/ contains the CSVs
ERROR: pandas import fails -> pip install pandas numpy scikit-learn
ERROR: MySQL table doesn't exist -> python manage.py migrate datasets
ERROR: UnicodeEncodeError on Windows -> set PYTHONIOENCODING=utf-8
ERROR: Memory on very large files -> chunksize is 200000; increase swap if needed
Pipeline not showing new files -> run 'python scripts/validate_datasets.py' first to generate statistics.json, then re-run the Django command

All numbers in this document are computed from the real files on disk. If a dataset file is missing, the pipeline reports MISSING, not a fabricated count.